

2016.0 RANGE ROVER (LG), 414-01

BATTERY, MOUNTING AND CABLES

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the battery system and operation, refer to the relevant Description and Operation section of the workshop manual REFER to: [Battery and Cables](#) (414-01 Battery, Mounting and Cables, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTES:

- Generic scan tools may not read the codes listed, or may read only five digit codes. Match the five digits from the scan tool to the first five digits of the seven digit code listed to identify the fault (the last two digits give additional information read by the manufacturer-approved diagnostic system).
- When performing electrical voltage or resistance tests, always use a digital multimeter (DMM) accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance, always take the resistance of the DMM leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.

1. Verify the customer concern

2. Visually inspect for obvious signs of mechanical or electrical damage

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Generator ▪ Drive belt ▪ Drive belt tensioner ▪ Generator pulley ▪ Check the security of the generator fixings	▪ Generator ▪ Battery ▪ Battery connections ▪ Starter motor ▪ Harnesses and connectors ▪ Fuses ▪ Charge warning indicator function ▪ Engine Control Module (ECM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
DTC B11DB-87 logged in Gateway Module	- Gateway Module connections insecure - Battery Monitoring System Control Module connections insecure - Battery Monitoring System Control Module fault	GO to Pinpoint Test A.
Battery power to vehicle interrupted	High resistance between battery terminals and clamps	GO to Pinpoint Test B.

DIAGNOSTIC PROCEDURES FOR DTC B11DB-87 (BATTERY MONITORING MODULE "A" - MISSING MESSAGE)

NOTES:

- These diagnostics only apply to vehicles fitted with a Gateway Module.
- If DTC B11DB-87 is flagged but a "charge system fault" warning / light is not present on the instrument cluster, then the likelihood is that the Battery Monitoring System Control Module is NOT the faulty part.

If a vehicle is presented with DTC B11DB-87 logged, follow the diagnostic routine below BEFORE replacing the Battery Monitoring System Control Module:

PINPOINT TEST A : BATTERY MONITORING SYSTEM CONTROL MODULE - FURTHER DIAGNOSTICS	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: CHECK FOR RELATED DTCS (B1412-87)	
	1 Using the manufacturer-approved diagnostic system, check the Gateway Module for related DTCS
	In addition to DTC B11DB-87, is DTC B1412-87(Quiescent Current Control Module - Missing Message) also logged? Yes Check the continuity of Gateway Module LIN_1 circuit and rectify as required. Then, GO to A2 . No GO to A3 .
A2: CLEAR DTCS AND RETEST. CHECK GATEWAY MODULE AND QUIESCENT CURRENT CONTROL MODULE	
	1 Using the manufacturer-approved diagnostic system, clear DTCS and retest the vehicle (vehicle must have the ignition on or engine running for a minimum of 20 seconds before retesting)
	Are DTCS still logged? Yes If ONLY B11DB-87 is logged, GO to A3 . If ONLY B1412-87 is logged, check the connections and operation of the Quiescent Current Control Module and rectify as required If BOTH DTCS are still present, check the connections and operation of the Gateway Module and rectify as required

	No No further action
A3: CHECK BATTERY MONITORING SYSTEM CONTROL MODULE INSTALLATION AND CONNECTIONS	
	1 Check that the Battery Monitoring System Control Module is seated correctly on battery post (the battery post should slightly protrude from the mounting ring). Adjust as required
	2 Check that the Battery Monitoring System Control Module power cable (thin wire from positive terminal) is securely connected to positive terminal and that the wiring is undamaged. Rectify as required
	3 Check the installation of the Battery Monitoring System Control Module connector for security and check the security of the cables into the connector. Rectify as required
	4 Disconnect the Battery Monitoring System Control Module and LIN circuit connectors and visually inspect the pins in both ends for damage. Rectify as required and resecure connectors
	5 NOTE: this step only applies to X250. Ensure that the earth cable fixing at location G4D178 is secure and torqued to 13.5Nm
	6 Using the manufacturer-approved diagnostic system, clear DTCs and retest the vehicle (vehicle must have the ignition on or engine running for a minimum of 20 seconds before retesting)
	Is DTC B11DB-87 still logged? Yes Replace the Battery Monitoring System Control Module No No further action

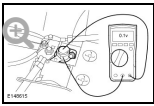
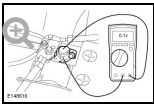
MIDTRONICS EXP-1080 USER GUIDE

NOTE:

This test checks for high resistance between the battery terminal and the battery clamp.

NOTE:

This test checks for high resistance between the battery terminal and the battery clamp.

PINPOINT TEST B : VOLTAGE DROP	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: GROUND CIRCUIT	
	1 Start the engine, turn on the following: ▪ (1) Air conditioning ▪ (2) Blower fan on full speed ▪ (3) Headlights on main beam ▪ (4) Heated screen - rear ▪ (5) Heated screen - front (if installed) ▪ (6) Heated seats (if installed)
	2 Connect the multimeter between the battery negative terminal and the battery clamp as shown in picture below (do not disconnect the battery at this stage)
	3 Set the multimeter to read DC voltage and record the reading
	Is reading equal to or below 0.1 volts? Yes GO to B2. No Switch all electrical loads and engine off, return the vehicle to an ignition off condition. Disconnect the battery negative clamp, clean clamp and terminal then reconnect and repeat test GO to B1.
B2: POWER CIRCUIT	
	1 Start the engine, turn on the following: ▪ (1) Air conditioning ▪ (2) Blower fan on full speed ▪ (3) Headlights on main beam

	▪ (4) Heated screen - rear ▪ (5) Heated screen - front (if installed) ▪ (6) Heated seats (if installed)
	2 Connect the multimeter between the battery positive terminal and the battery clamp as shown in picture below (do not disconnect the battery at this stage)
	3 Set the multimeter to read DC voltage and record the reading
	Is reading equal to or below 0.1 volts? Yes Carry out Midtronics battery test procedure No Switch all electrical loads and engine off, return the vehicle to an ignition off condition. Disconnect the battery positive clamp, clean clamp and terminal then reconnect and repeat test GO to B2.

The following steps must be carried out to ensure correct operation of the EXP-1080 battery tester during the battery test procedure

CHECKS	ACTION
Battery fluid leakage, check for battery fluid leaks or damage to the battery casing	NOTE: If visible damage to the case is evident do not return battery under warranty. Replace the battery if there is any battery fluid leaks evident
Battery vent pipe routing	Check for routing, ensure there are no kinks
EXP-1080 fly lead, condition of clamps	Clean or replace as required
EXP-1080 fly lead connection	Confirm secure connection

NOTE:

The Midtronics EXP-1080 battery tester is suitable for testing flooded and absorbed glass mat (AGM) type batteries including Primary and Secondary batteries.

Testing a Battery

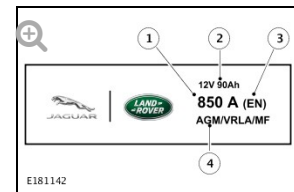
Using the Midtronics EXP-1080 battery tester, the following test procedure will confirm the serviceability of the battery (see Completing a Battery Test)

Battery Label Example

NOTE:

All AGM batteries are marked with AGM. Flooded batteries have no reference to being Flooded.

- 1. Battery Rating Units CCA (Cold Cranking Amps)
- 2. Battery voltage and Battery Ah rating
- 3. Battery rated units (battery standard EN or SAE)
- 4. Battery type (battery technology, AGM or Flooded)



Battery Test Types

The Midtronics EXP-1080 battery tester has three types of battery test available to the technician to select:

Battery Test

- 1. The BATTERY TEST should be used on any battery that has started its warranty life cycle. The battery is in use and fitted to a vehicle registered to an owner

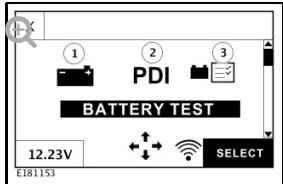
PDI / Storage

- 2. The PDI / STORAGE test should be used on any battery that has not yet been entered into the warranty life cycle. The battery is fitted to a NEW vehicle, but the vehicle has not yet been sold/registered to an owner

Battery Storage

- 3. This BATT. STORAGE test should be used on any battery that has not yet been entered into the warranty life cycle. The battery is not in use and is a Parts Stock battery and has not yet been fitted to a vehicle

Completing a Battery Test

1	Connect the fly-lead to the Midtronics EXP-1080 battery tester	
2	Connect the fly-leads to the battery terminals, black lead to the negative terminal, red lead to the positive terminal and confirm the connections are secure	
3	The battery tester will power ON automatically when connected to a battery	
4	MAIN MENU - Select the correct BATTERY TEST using the arrow keys on the battery tester panel (see Battery Test Types) 1. BATTERY TEST 2. PDI / STORAGE 3. BATT. STORAGE NOTE: When BATT. STORAGE is selected, the technician must enter IDENTIFICATION data or a purchase order reference number for the battery being tested. The battery tester will then move on to BATTERY TYPE (Step 8). Select NEXT to continue	

5 BRAND SELECTION

NOTE:

This step is required for BATTERY TEST and PDI/STORAGE only.

Select the correct brand of the vehicle battery being tested using the scroll arrow keys on the battery tester panel

Select NEXT to continue

BRAND SELECTION

1 ☒ JAGUAR
2 ☐ LAND ROVER

BACK NEXT

E181143

6 VIN

NOTES:

- This step is required for BATTERY TEST and PDI/STORAGE only.
- Letters SAL or SAJ may be different than displayed on the EXP-1080 if the vehicle is manufactured outside the United Kingdom, this will not effect the battery testing.

Enter the 4th and 5th digit followed by the last 6 of the VIN using the alphanumeric key pad on the battery tester panel

Select NEXT to continue

VIN

ENTER DIGIT 4 AND 5 AND
LAST 6 OF VIN

XXX XXXXXX

BACK NEXT

E181144

7 BATTERY LOCATION

NOTES:

- This step is required for BATTERY TEST and PDI/STORAGE only.
- Selecting the correct battery details at this point is critical to getting the correct Battery Test Results.

Select IN VEHICLE if the battery is tested connected to the vehicle

BATT. LOCATION

1 ☒ IN VEHICLE
2 ☐ OUT OF VEHICLE

BACK NEXT

E164366

NOTE:

Make sure the Ignition is OFF and the vehicle is powered down with the modules entering 'sleep mode' before commencing with the battery test.

Select OUT OF VEHICLE only if the battery is electrically disconnected from the vehicle or on a bench

Select NEXT to continue

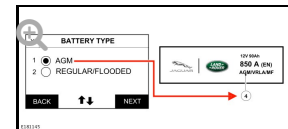
8 BATTERY TYPE

NOTES:

- All AGM batteries are marked with AGM. Flooded batteries have no reference to being Flooded.
- Selecting the correct battery details at this point is critical to getting the correct Battery Test Results.

Select the correct battery type (see Battery Label Example, Label 4) using the scroll arrow keys on the battery tester panel

Select NEXT to continue



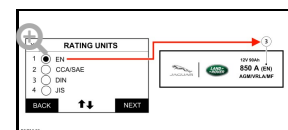
9 RATING UNITS

NOTE:

Selecting the correct battery details at this point is critical to getting accurate Battery Test Results.

Select the correct battery rating from the battery label in brackets (see Battery Label Example, Label 3) using the scroll arrow keys on the battery tester panel

Select NEXT to continue



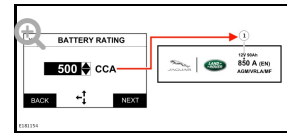
10 BATTERY RATING

NOTE:

Selecting the correct battery details at this point is critical to getting accurate Battery Test Results.

Select the correct CCA rating (see Battery Label Example, Label 1) using the scroll arrow keys, or enter the rating with the numeric key pad on the battery tester panel

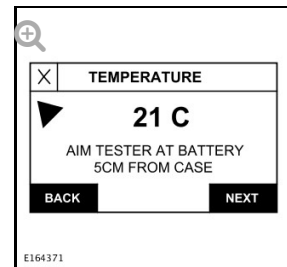
Select NEXT to continue



11 TEMPERATURE

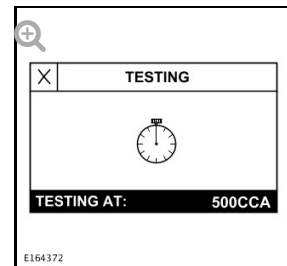
Aim the temperature sensor towards the battery casing (maintain a distance of 5 cm)

Select NEXT to continue



12 TESTING

The battery tester screen will now display clock hands rotating, the battery tester will automatically advance when the test has completed



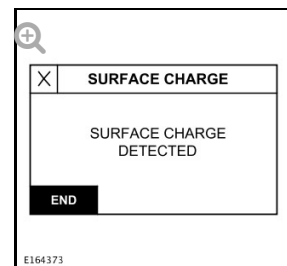
13 SURFACE CHARGE

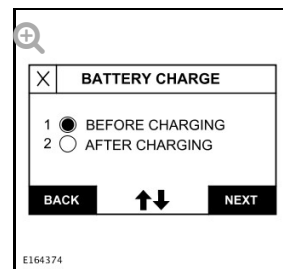
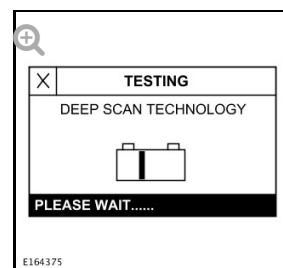
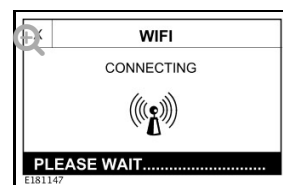
NOTE:

This step will take place automatically only when required.

The SURFACE CHARGE test is an additional step required if the battery voltage is above 12.55v with a low CCA measured

- Follow the battery tester on screen instructions
- 1. Turn the ignition ON (position 2)
- 2. Turn ON the headlights (high beam) until the battery tester shows- Turn OFF headlights



	3. Turn the ignition OFF (position 0)	
14	BATTERY CHARGE NOTES: - This step will take place automatically only when required. - To support a Warranty Claim, TEST CODE details for BEFORE CHARGING and AFTER CHARGING must be supplied in the technical comments box of the Warranty Claim. Using the scroll arrow keys on the battery tester panel: Select BEFORE CHARGING if the battery has not been connected to a recommended mains charger Select AFTER CHARGING if the battery has been connected to a recommended mains charger for the recommended time shown on the result screen Select NEXT to continue	
15	DEEP SCAN NOTE: This step will take place automatically only when required. The battery tester will display and complete an automated DEEP SCAN test only if required, then automatically advance when the DEEP SCAN test has completed	
16	CONNECTING to Wi-Fi	

NOTES:

- A warning message is displayed if the battery tester is unable to connect to a Wi-Fi network. Locate WIFI SETTINGS in the UTILITY MENU and follow the on screen instructions.
- If the battery tester is not fitted with a Wi-Fi pod, the battery tester will move on to RESULTS (Step 19).

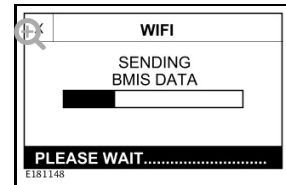
The battery tester will now use the A331 Wi-Fi pod (if installed) to connect to the local Wi-Fi network and connect to the Battery Management Information Service (BMIS) server

17 SENDING BMIS DATA

NOTE:

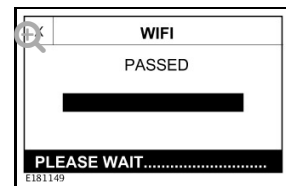
A warning message is displayed if the battery tester is unable to connect to the BMIS server. Your dedicated IT department may be required to complete all or some of the set up procedure.

The battery test result data is now being transferred to the BMIS server



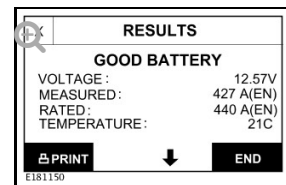
18 BMIS DATA PASSED to SERVER

The battery test result data has been successfully saved to the BMIS server



19 RESULTS

Review the battery test results and complete the appropriate actions (see Battery Tester Results Table)

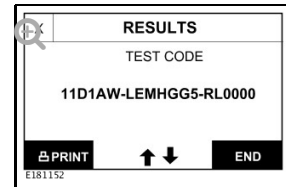


20 TEST CODE

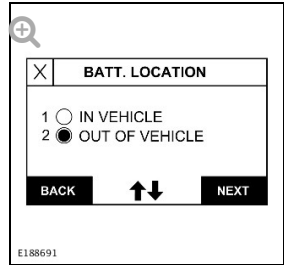
NOTE:

To support a Warranty Claim, TEST CODE details for BEFORE CHARGING and AFTER CHARGING must be supplied in the technical comments box of the Warranty Claim.

From the RESULTS display, view the TEST CODE using the scroll arrow keys on the battery tester panel

**Battery Tester Results Table**

RESULTS	ACTION	
GOOD BATTERY	Return the battery to service	
CHARGE AND RE-TEST	NOTE: To support a Warranty Claim, TEST CODE details for Before Charging and After Charging must be supplied in the technical comments box of the Warranty Claim.	A screenshot of a battery tester's LCD screen. It says 'RESULTS' with a small 'X' icon. Below is 'CHARGE & RETEST'. The data shown is: VOLTAGE: 12.57V, MEASURED: 390 A(EN), RATED: 440 A(EN), TEMPERATURE: 21C. At the bottom are 'PRINT', scroll arrow, and 'END' buttons. Model number 'E181151' is at the bottom.
	When the battery test result equals CHARGE & RETEST, view the TIME TO CHARGE screen using the scroll arrow keys on the Battery Tester panel	A screenshot of a battery tester's LCD screen. It says 'RESULTS' with a small 'X' icon. Below is 'TIME TO CHARGE (smart charger)'. The data shown is: 150 MINUTES @ 25A, 75 MINUTES @ 50A. At the bottom are 'PRINT', scroll arrow, and 'END' buttons. Model number 'E164378' is at the bottom.
	Follow the 50AMP charge time for all vehicles, except Defender which can be charged with a 25 Amp charger	
	Follow the recommended action in the test results	
	If the second battery test result equals CHARGE & RETEST the battery must be replaced	
REPLACE BATTERY OR BAD CELL-REPLACE	1. Repeat the test with the battery disconnected from the vehicle. Ensure OUT OF VEHICLE is selected as the Battery Location for the repeat test 2. Follow the instructions as per the test result	

	For a repeat REPLACE BATTERY or BAD-CELL test result, DO NOT ATTEMPT TO RECHARGE THE BATTERY, REPLACE THE BATTERY	
FROZEN BATTERY	Allow the battery to thaw naturally in workshop conditions and re-test	
UNABLE TO DO TEST	Disconnect the battery from the vehicle and re-test	

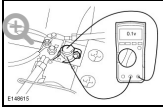
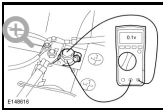
Flooded Battery Care Point

If the vehicle is equipped with a flooded battery, ensure the replacement battery is a flooded battery of the same specification (cold cranking amperage (CCA), battery standard (EN/SAE) / amp hour rating (Ah)) as the original battery

Under no circumstances should you fit a flooded battery to a vehicle that originally had an AGM battery, unless formally instructed by Jaguar/Land Rover

AGM Battery Care Point

If the vehicle is equipped with an absorbed glass mat (AGM) battery, ensure the replacement battery is a AGM battery of the same specification (cold cranking amperage (CCA), battery standard (EN/SAE) / amp hour rating (Ah)) as the original battery. Replacing an AGM battery for a flooded type battery is likely to result in reduced battery life and should not be carried out, unless formally instructed by Jaguar/Land Rover

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: GROUND CIRCUIT	
	1 Start the engine, turn on the following: ▪ (1) Air conditioning ▪ (2) Blower fan on full speed ▪ (3) Headlights on main beam ▪ (4) Heated screen - rear ▪ (5) Heated screen - front (if installed) ▪ (6) Heated seats (if installed)
	2 Connect the multimeter between the battery negative terminal and the battery clamp as shown in picture below (do not disconnect the battery at this stage)
	3 Set the multimeter to read DC voltage and record the reading
	Is reading equal to or below 0.1 volts? Yes GO to C2. No Switch all electrical loads and engine off, return the vehicle to an ignition off condition. Disconnect the battery negative clamp, clean clamp and terminal then reconnect and repeat test
C2: POWER CIRCUIT	
	1 Start the engine, turn on the following: ▪ (1) Air conditioning ▪ (2) Blower fan on full speed ▪ (3) Headlights on main beam ▪ (4) Heated screen - rear ▪ (5) Heated screen - front (if installed) ▪ (6) Heated seats (if installed)
	2 Connect the multimeter between the battery positive terminal and the battery clamp as shown in picture below (do not disconnect the battery at this stage)

	Is reading equal to or below 0.1 volts? Yes Carry out Midtronics battery test procedure No Switch all electrical loads and engine off, return the vehicle to an ignition off condition. Disconnect the battery positive clamp, clean clamp and terminal then reconnect and repeat test
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The following steps must be carried out to ensure correct operation of the GRX-3080 diagnostic battery charger during the battery test procedure

CHECKS	ACTION
Battery fluid leakage, check for battery fluid leaks or damage to the battery casing	NOTE: If visible damage to the case is evident do not return battery under warranty. Replace the battery if there is any battery fluid leaks evident
Battery vent pipe routing	Check for routing, ensure there are no kinks
GRX-3080 fly lead, condition of clamps	Clean or replace as required
GRX-3080 fly lead connection	Confirm secure connection

NOTE:

The GRX-3080 diagnostic battery charger is suitable for testing flooded and absorbed glass mat (AGM) type batteries including Primary and Secondary batteries

Testing a Battery

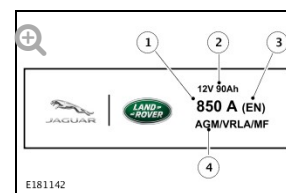
Using the GRX-3080 diagnostic battery charger, the following test procedure will confirm the serviceability of the battery (see Completing a Battery Test)

Battery Label Example

NOTE:

All AGM batteries are marked with AGM. Flooded batteries have no reference to being Flooded.

- 1. Battery Rating Units CCA (Cold Cranking Amps)
- 2. Battery voltage and Battery Ah rating
- 3. Battery rated units (battery standard EN or SAE)
- 4. Battery type (battery technology, AGM or Flooded)



GRX-3080 Diagnostic Battery Charger Display Keypad

The GRX-3080 diagnostic battery charger has an integrated display and keypad, the display guides the user via onscreen navigation aids, directions and messages. The illustration (below) shows how the elements on the screen relate to the keypad and details the functions of the various keys:

KEY	DESCRIPTION	GRX-3080 USER INTERFACE
1	■ VOLTAGE – Battery voltage displayed when first connected	
2	■ SOFT KEYS - Pressing the two soft keys linked to the bottom of the screen will perform the functions displayed above them. The functions change depending on the menu and test process	
3	■ ARROW KEYS - Press the ARROW keys to scroll through numerical values and navigate through menus and icons	
4	■ STOP BUTTON – Press STOP key anytime to stop the active mode	
5	■ Title Bar – The title bar shows the name of the current menu	
6	■ Selection Area – Contains selectable icons or dialogue boxes that	

	display information and required response	
7	▪ Menu Screen Arrows - When displayed in menu screens, the menu screen arrows shows which ARROW key on the keypad to press to display other icons or screens. The Up/Down Menu Screen Arrows indicate when to press the UP or DOWN keys to display the screens above and below the current screen ▪ The left/right menu screen arrows tell you when to use the LEFT or RIGHT keys to select an icon ▪ When displayed under a list of options, the menu screen arrows show you which keypad arrow to press to highlight a character or item in a list	
8	▪ Alphanumeric Keypad - The alphanumeric keypad enters letters and numbers for the PAG code, VIN and Battery storage entry. It also can be used to customise personnel settings	

Battery Test Types

Before performing any battery diagnostic on a vehicle, make sure the ignition is OFF and the vehicle is powered down, with the modules entering 'sleep mode' before commencing and with the hazard switch illumination extinguished

The Midtronics GRX-3080 JLR diagnostic battery charger has three types of battery test (Diagnostic Charging, PDI/ Storage and Battery Storage), one battery support function (Battery Storage) and one manual charging option available (Battery Support Power Supply) for the technician to select:

DIAGNOSTIC CHARGING

- 1. The DIAGNOSTIC CHARGING test needs to be used on any battery that has started its warranty life cycle. The battery is in use and fitted to a vehicle registered to an owner. The GRX Battery Diagnostic Station will determine the internal condition of a battery before attempting to apply a charge to it. If required, it will then charge and assess the battery until the battery condition is established and a test result is given

PDI / STORAGE

- 2. The PDI / STORAGE test needs to be used on any battery that has not yet been entered into the warranty life cycle. The battery is fitted to a NEW vehicle, but the vehicle has not yet been sold/registered to an owner. The GRX Battery Diagnostic Station will determine the internal condition of a battery before attempting to apply a charge to it. If required, it will then charge and assess the battery until the battery condition is established and a test result is given

BATTERY STORAGE

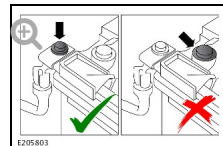
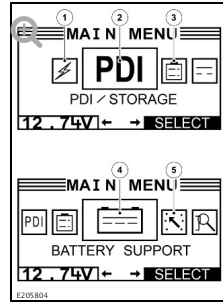
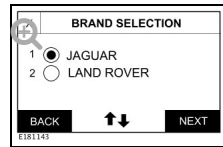
- 3. The BATT. STORAGE test should be used on any battery that has not yet been entered into the warranty life cycle. The battery is not in use and is a Parts Stock battery and has not yet been fitted to a vehicle. The GRX Battery Diagnostic Station will determine the internal condition of a battery before attempting to apply a charge to it. If required, it will then charge and assess the battery until the battery condition is established and a test result is given

BATTERY SUPPORT Power Supply

- 4. The BATTERY SUPPORT Power Supply function needs to be used whenever the vehicle comes in to the workshop for regular maintenance or vehicle module updates. Updating modules can take up to several hours and during that period a lot of current will be depleted from the battery. To support the battery during this process use the BATTERY SUPPORT function. When working on a vehicle, it is recommended that the GRX-3080 is connected and switched to the BATTERY SUPPORT function to ensure that, when the work is completed, the battery's state of charge is maintained at a healthy level

Completing a Battery Test

1	Connect the fly-lead to the Midtronics GRX-3080 JLR battery charger	
2	Connect the fly-leads to the battery terminals, black lead to the negative terminal, red lead to the positive terminal and	

	confirm the connections are secure	
3	The battery tester will power ON automatically when connected to a battery	
4	MAIN MENU - Select the correct BATTERY TEST using the arrow keys on the battery tester panel (see Battery Test Types above) 1. DIAGNOSTIC CHARGING. Go to step 5 2. PDI / STORAGE. Go to step 5 3. BATT. STORAGE. Go to step 5 4. BATTERY SUPPORT. Go to step 20 5. RECOVERY CHARGING. Go to step 21 NOTE: When BATT. STORAGE is selected, the technician must enter IDENTIFICATION data or a purchase order reference number for the battery being tested. The battery tester will then move on to BATTERY TYPE (Step 9) Select NEXT to continue	
5	BRAND SELECTION NOTE: This step is required for DIAGNOSTIC CHARGING and PDI/STORAGE only Select the correct brand of the vehicle battery being tested using the scroll arrow keys on the battery tester panel Select NEXT to continue	
6	VIN	

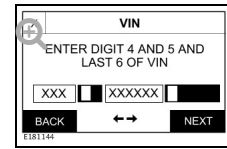
NOTE:

This step is required for DIAGNOSTIC CHARGING and PDI/STORAGE only

Enter the 4th and 5th digit followed by the last 6 of the VIN using the alphanumeric key pad on the battery tester panel

When BATT. STORAGE is selected, the technician must enter IDENTIFICATION data or a purchase order reference number for the battery being tested

Select NEXT to continue

**7 BATTERY LOCATION****NOTES:**

- This step is required for DIAGNOSTIC CHARGING and PDI/STORAGE only
- Selecting the correct battery details at this point is critical to getting the correct Battery Test results

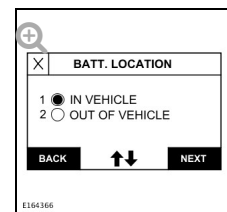
Select IN VEHICLE if the battery is tested connected to the vehicle

NOTE:

Make sure the ignition is OFF and the vehicle is powered down with the modules entering 'sleep mode' before commencing with the battery test

Select OUT OF VEHICLE only if the battery is electrically disconnected from the vehicle or on a bench

Select NEXT to continue

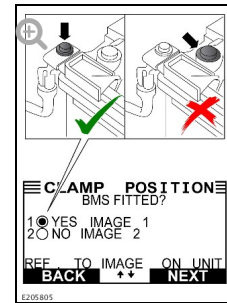
**8 BATTERY MONITORING SYSTEM (BMS) INFORMATION**
(Required if 'IN VEHICLE' has been selected)

NOTE:

'IMAGE' refers to the sticker that is located on top of the GRX-3080 JLR

Check to make sure the negative clamp is connected as in the picture on top of the GRX-3080 JLR diagnostic battery charger with a tick and, if so, select YES

Select NEXT to continue



9 BATTERY TYPE

NOTES:

- All AGM batteries are marked with AGM. Flooded batteries have no reference to being Flooded
- Selecting the correct battery details at this point is critical to getting the correct Battery Test results

Select the correct battery type (see Battery Label example, Label 4) using the scroll arrow keys on the battery tester panel

Select NEXT to continue



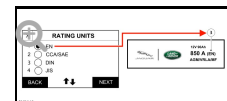
10 RATING UNITS

NOTE:

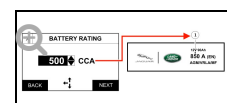
Selecting the correct battery details at this point is critical to getting the correct Battery Test results

Select the correct battery rating from the battery label in brackets (see Battery Label Example, Label 3) using the scroll arrow keys on the battery tester panel

Select NEXT to continue



11 BATTERY RATING



NOTE:

Selecting the correct battery details at this point is critical to getting the correct Battery Test results

Select the correct CCA rating (see Battery Label Example, Label 1) using the scroll arrow keys, or enter the rating with the numeric key pad on the battery tester panel

Select NEXT to continue

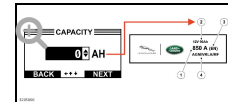
12 CAPACITY

NOTE:

Selecting the correct battery details at this point is critical to getting the correct Battery Test results

Select the correct CAPACITY rating (see Battery Label Example, Label 2) using the scroll arrow keys, or enter the rating with the numeric key pad on the battery tester panel

Select NEXT to continue



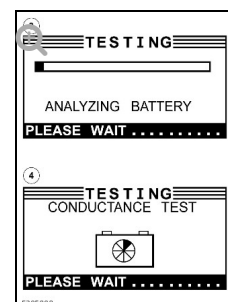
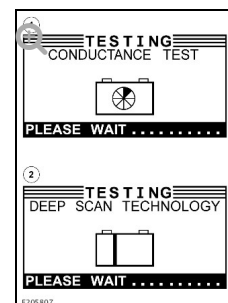
13 TESTING

- There are four automated screens within the testing, they will display sequentially with no user input required:
- 1. Conductance
- 2. Deep Scan Technology (If required, see note below)
- 3. Analysis
- 4. Conductance

NOTE:

The battery tester will display and complete an automated DEEP SCAN test only if required, then automatically advance when the DEEP SCAN test has completed

The GRX-3080 JLR diagnostic battery charger will automatically advance when the test has completed



14 DIAGNOSING

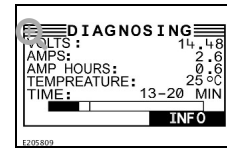
- Once the GRX-3080 JLR diagnostic battery charger determines that the battery is healthy, needs charging, and is safe to charge, it proceeds
- During the charging session, the GRX-3080 JLR diagnostic battery charger provides updates of the charging voltage, charging current, remaining time to charge and the amount of charge replenished into the battery in amp-hours
- The GRX-3080 JLR diagnostic battery charger continues to test the battery throughout the charge cycle and may determine at some point that the battery needs to be replaced even though the remaining time to charge is not completed

NOTE:

The actual time needed to charge the battery might be less than the estimated time depending on the battery's charge acceptance and condition during the charge

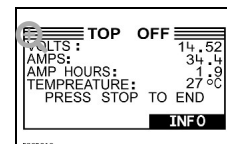
- Before and during the charge cycle the battery is tested with both Conductance and a Load Bounce
- The GRX-3080 JLR diagnostic battery charger uses the first portion of the total charge cycle to closely monitor the acceptance of charge current and the energy going in to the battery. During this mode, the charger continuously analyses the battery to make a decision as quickly as possible

Diagnostic mode will automatically move to next steps



15 TOP OFF

- This mode occurs automatically at the end of the diagnostic charge cycle to allow the battery to reach 100% state of charge. TOP OFF mode ends when the battery's acceptance of the charge has reached its limit



NOTE:

Do not stop the TOP OFF charge cycle as this will reduce the amount of charge the battery has received

- TOP OFF mode will automatically move to the result screen

16 CONNECTING TO WIFI

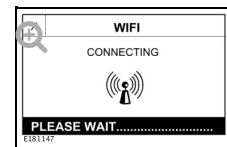
- This mode occurs automatically at the end of the TOP OFF mode

NOTES:

- A warning message is displayed if the battery tester is unable to connect to a Wi-Fi network. Locate WIFI SETTINGS in the UTILITY MENU and follow the on screen instructions
- If the battery tester is not fitted with a Wi-Fi pod, the battery tester will move on to RESULTS (Step 19)

The Wi-Fi pod will connect to the local Wi-Fi network to access and upload information to the Battery Management Information Service (BMIS) server

CONNECTING TO WIFI mode will automatically move to the result screen



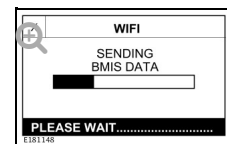
17 SENDING BMIS DATA

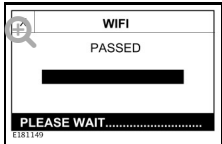
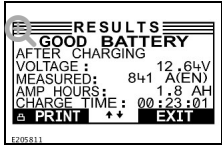
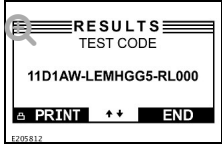
- This mode occurs automatically at the end of the CONNECTING TO WIFI mode

NOTE:

A warning message is displayed if the battery tester is unable to connect to the BMIS server. Your local IT department may be required to complete all or some of the set up procedure

The battery test result data is transferred to the BMIS server



18	BMIS DATA PASSED to SERVER	
	The battery test result data has been successfully saved to the BMIS server	
19	RESULTS	 
	From the RESULTS display, view the TEST CODE using the scroll arrow keys on the battery tester panel Select PRINT for your records or exit to go back to the main menu screen	

Battery Tester Results Table

RESULTS	ACTION
GOOD BATTERY	Return the battery to service
REPLACE BATTERY	NOTE: A 'REPLACE BATTERY' result may also mean a poor connection between the battery cables on the vehicle and the battery. Check connections and rectify as required - Ensure the volt drop test has been completed (see PINPOINT TEST A: VOLTAGE DROP ACROSS VEHICLE HARNESS AND BATTERY TERMINAL above) - REPLACE THE BATTERY
FROZEN BATTERY	Allow the battery to thaw naturally in workshop conditions and retest
UNABLE TO DO TEST	Disconnect the battery from the vehicle and retest

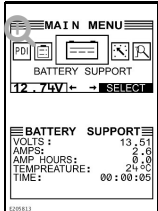
Flooded Battery Care Point

If the vehicle is equipped with a flooded battery, ensure the replacement battery is a flooded battery of the same specification (cold cranking amperage (CCA), battery standard (EN/SAE) / amp hour rating (Ah)) as the original battery

Under no circumstances should you fit a flooded battery to a vehicle that originally had an AGM battery, unless formally instructed by Jaguar/Land Rover

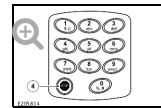
AGM Battery Care Point

If the vehicle is equipped with an absorbed glass mat (AGM) battery, ensure the replacement battery is a AGM battery of the same specification (cold cranking amperage (CCA), battery standard (EN/SAE) / amp hour rating (Ah)) as the original battery. Replacing an AGM battery for a flooded type battery is likely to result in reduced battery life and should not be carried out, unless formally instructed by Jaguar/Land Rover

20	BATTERY SUPPORT ▪ The Battery Support function is used whenever the vehicle comes in to the workshop for regular maintenance or module downloads ▪ Re-flashing an ECU can take up to several hours and during that period a lot of current can be drawn from the battery. To support the battery during this process, use the Battery Support function ▪ When working on a vehicle, it is recommended that the GRX-3080 is connected and switched to the BATTERY SUPPORT function to ensure that, when the work is completed, the battery's state of charge is maintained at a healthy level ▪ Make sure the GRX-3080 JLR diagnostic battery charger is connected (as in steps 1 and 2 from the MAIN Menu above), highlight the Battery Support icon and press the SELECT key ▪ The GRX-3080 JLR diagnostic battery charger will immediately begin to supply power and the BATTERY SUPPORT screen will display a list of real-time battery data	

NOTE:

BATTERY SUPPORT mode will not automatically move to next step. When battery support is no longer required, press the STOP button on the display panel (see Label 4)

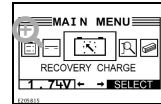


Select STOP when battery support is not required

21 RECOVERY CHARGE

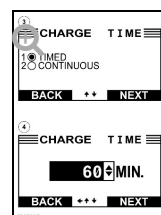
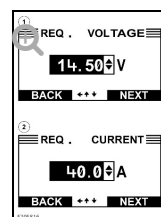
NOTE:

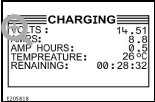
Only use the RECOVERY CHARGE mode when the battery voltage is 9 volts or below. This will charge the battery for 1 hour. When the time has elapsed, use either DIAGNOSTIC CHARGING or PDI/STORAGE mode to complete the analysis of the battery (see the section on Battery Test Types (above) for guidance on when to use which mode)



- **Follow steps 7 to 11 (as detailed above) to enter battery details, including:**
- BATTERY LOCATION (Step 7)
- BMS INFORMATION (If "IN VEHICLE" selected) (Step 8)
- BATTERY TYPE (Step 9)
- RATING UNITS (Step 10)
- BATTERY RATING (Step 11)

- **Next, enter the following settings:**
- **1. REQ. VOLTAGE -**
 - For both AGM and Flooded Batteries, use the numeric keypad to enter 14.5 volts
- **2. REQ. CURRENT**
 - For AGM Batteries, use the numeric keys to enter 40.0 amps
 - For Flooded Batteries, use the numeric keys to enter 50.0 amps
- **3. CHARGE TIME (1)**
 - Select the "Timed" option
- **4. CHARGE TIME (2)**
 - Use the alphanumeric keypad and type 60 MIN



Select NEXT to continue	
NOTE: During the charging session, the GRX-3080 JLR diagnostic battery charger provides updates of the charging voltage, current, remaining time to charge and the amount of amp hours the battery received ■ The GRX-3080 JLR diagnostic battery charger will not continue to test the battery throughout the Recovery Charge ■ When the hour has finished the GRX-3080 JLR, diagnostic battery charger will give 4 audible beeps to confirm the mode has completed	
Select EXIT to complete Recovery Charging	

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.